

CSE3156 _DBMS Question Bank

List the limitations/disadvantages of DBMS.	2 Marks
List the major functions performed by DBA.	2 Marks
Differentiate between weak entity and strong entity.	2 Marks
Explain various Data types used in SQL.	2 Marks
What is nested query in SQL? Explain with a suitable SQL query.	2 Marks
Explain ON DELETE CASCADE constraint with suitable SQL query.	2 Marks
Explain super key and candidate key.	2 Marks
Define Functional Dependency.	2 Marks
Define the concept of a database transaction with an example.	2 Marks
Explain briefly the meaning of serializability of transactions.	2 Marks
List the benefits of database approach.	2 Marks
List and explain the different types of database end users.	2 Marks
Analyze the output of the following statement? $\sigma_{\text{subject} = \text{"database"}}(\text{Books})$	2 Marks
Explain various data types used in SQL.	2 Marks
What is Natural Join in SQL? Explain with a simple SQL query.	2 Marks
What is Embedded SQL? Explain with a simple code snippet.	2 Marks
What is Primary Key? Explain the desirable characteristics of it.	2 Marks
Define trivial and non-trivial functional dependencies.	2 Marks
What are concurrent transactions? Explain the need of concurrency control in brief.	2 Marks
Write about SQL support for transactions.	2 Marks
Explain the significance of pre-compiler in DBMS environment.	2 Marks
Define the terms: database schema and database state.	2 Marks
Compare weak and strong entities with suitable example.	2 Marks
Explain various Data types used in SQL.	2 Marks
Write an SQL Query to display second highest salary from an EMPLOYEE table.	2 Marks

CSE3156 _DBMS Question Bank

Differentiate between DROP and DELETE SQL statements.	2 Marks																																			
What are prime and non-prime attributes? Explain with an example.	2 Marks																																			
Explain partial functional dependency and full functional dependency with an example.	2 Marks																																			
Define the commit point of a transaction.	2 Marks																																			
What are the three problems that any concurrency control mechanism must address? Explain in brief.	2 Marks																																			
Design an E-R diagram for the PRESIDENCY UNIVERSITY database, which is being designed to manage information about departments, courses, instructors, and students. The university has multiple departments. Each department is identified by a unique Department ID, has a name, and a building location. Each department offers multiple courses, but a course is offered by only one department. A Course has a Course ID, title, and credit hours. An Instructor is assigned to exactly one department but can teach multiple courses. Each instructor has a unique ID, name, and email. A Student can enroll in multiple courses. Each student has a unique Student ID, name, and email. A course can be taken by many students, and each student can take many courses. We also need to keep track of which instructor teaches which course.	10 Marks																																			
Differentiate among NATURAL JOIN, EQUI-JOIN, LEFT OUTER JOIN, and RIGHT OUTER JOIN Relational Algebra operations using the EMPLOYEE and DEPARTMENT relations given below. EMPLOYEE <table border="1"><thead><tr><th>ENO</th><th>NAME</th><th>DOB</th><th>GENDER</th><th>DCODE</th></tr></thead><tbody><tr><td>12345</td><td>HAMEN</td><td>24-MAR-2001</td><td>M</td><td>201</td></tr><tr><td>12346</td><td>VINI</td><td>12-MAR-2001</td><td>F</td><td>202</td></tr><tr><td>12347</td><td>ANI</td><td>11-JAN-1999</td><td>F</td><td></td></tr><tr><td>12348</td><td>PETER</td><td>14-FEB-2001</td><td>M</td><td></td></tr></tbody></table> DEPARTMENT <table border="1"><thead><tr><th>DCODE</th><th>DNAME</th></tr></thead><tbody><tr><td>201</td><td>COMPUTER SC</td></tr><tr><td>202</td><td>INFN SC</td></tr><tr><td>203</td><td>CIVIL</td></tr><tr><td>204</td><td>MECHANICAL</td></tr></tbody></table>	ENO	NAME	DOB	GENDER	DCODE	12345	HAMEN	24-MAR-2001	M	201	12346	VINI	12-MAR-2001	F	202	12347	ANI	11-JAN-1999	F		12348	PETER	14-FEB-2001	M		DCODE	DNAME	201	COMPUTER SC	202	INFN SC	203	CIVIL	204	MECHANICAL	10 Marks
ENO	NAME	DOB	GENDER	DCODE																																
12345	HAMEN	24-MAR-2001	M	201																																
12346	VINI	12-MAR-2001	F	202																																
12347	ANI	11-JAN-1999	F																																	
12348	PETER	14-FEB-2001	M																																	
DCODE	DNAME																																			
201	COMPUTER SC																																			
202	INFN SC																																			
203	CIVIL																																			
204	MECHANICAL																																			
What is a View? Explain the problems associate with updating the Views. Demonstrate with simple SQL queries.	5 Marks																																			

CSE3156 _DBMS Question Bank

Explain the following SQL commands/Keywords: i. REVOKE ii. GRANT iii. COMMIT iv. ROLLBACK v. DISTINCT	5 Marks
Consider the following relations for a company database Application. EMPLOYEE (<u>ENO</u>, ENAME, SEX, DOB, DOJ, BASIC_PAY, DEPTNO) DEPARTMENT (<u>DEPTNO</u>, DNAME) PROJECT (<u>PROJNO</u>, PNAME, DEPTNO) WORKS_ON (ENO, PROJNO, HOURS) Implement the SQL queries for the following requirements. i. Calculate the numbers of male and female employees. ii. List the names of employees who are working for CSE department. iii. Count the number of employees who are working on JAVA project. iv. List the employees who are not working on any projects. Create a view to show the names of all the employees along with their department names.	10 Marks
What is a Database Trigger? Create a simple UPDATE trigger using SQL on a table EMPLOYEE.	5 Marks
What is a stored procedure? Create a stored procedure to add two numbers and print the result.	5 Marks
Consider the relation schema given below: EMPLOYEE (EMPNO, NAME, OFFICE, AGE) BOOK (ISBN, TITLE, AUTHOR, PUBLISHER) LOAN (EMPNO, ISBN, DATE) Implement the SQL queries for the following requirements. i. Find the names of employees who have borrowed a book published by McGraw-Hill. ii. List the names of the employees who have not borrowed any books. iii. List the names of the employees who have borrowed more than two books. iv. Retrieve the total number of books present in the library. Generate a report containing EMPNO, NAME, ISBN, AUTHOR, PUBLISHER, and BORROWED DATE.	10 Marks

CSE3156 _DBMS Question Bank

Explain the informal design guidelines used in relational schema design. Why are these guidelines important, and how do they help in ensuring an effective database design? Illustrate your answer with suitable examples.	5 Marks
What is multivalued dependency? Explain with an example.	5 Marks
Consider the Relation R (A, B, C, D, E, F) and Functional Dependencies: {AB->C, A->D, B->E, C->F}. Normalize the given relation to its highest normal form. Explain the steps.	10 Marks
Explain the Armstrong's Rules of Inference.	5 Marks
What is a lossless decomposition? Explain with an example.	5 Marks
What is normalization? Explain 1NF, 2NF and 3NF with real world examples.	10 Marks
Show the different states of a transaction in detail with the help of a well-labeled state diagram.	5 Marks
Explain the concept of concurrency control in transaction management. Analyze why it is essential in multi-user database environment?	5 Marks
What are two-phase locking protocols? How does it guarantee serializability? Explain with an example.	10 Marks
List and explain the ACID properties of a database transaction.	5 Marks
Explain Time stamp ordering algorithm.	5 Marks
Write a short note on DBMS i. Recovery manager ii. System log file.	10 Marks
Explain the different types of DBMS users.	5 Marks
Explain the three-schema Architecture. What is the difference between logical data independence and physical data independence?	5 Marks
Explain in detail the components of DBMS environment with a block diagram.	10 Marks
Explain various unary operators used in Relational Algebra with proper examples.	5 Marks
Construct E-R diagram of the bank. It provides different kinds of bank accounts and loans. It operates number of branches.	5 Marks

CSE3156 _DBMS Question Bank

Consider the following relations Suppliers(sID, sName, address) Parts(pID, pName, colour) Catalog(sID, pID, price) Write the relational algebra for the following requirements: - Find the names of all red parts. - Find all prices for parts that are red or green. - Find the sIDs of all suppliers who supply a part that is red or green Find the names of all suppliers who supply a part that is red or green.	10 Marks
What is a View? How can it be created and dropped? Demonstrate with simple SQL queries.	5 Marks
Explain the following SQL commands/Keywords: - REVOKE - GRANT - COMMIT - ROLLBACK - DISTINCT	5 Marks
Consider the following relations for a company database Application. EMPLOYEE (<u>ENO</u>, ENAME, SEX, DOB, DOJ, BASIC_PAY, DEPTNO) DEPARTMENT (<u>DEPTNO</u>, DNAME) PROJECT (<u>PROJNO</u>, PNAME, DEPTNO) WORKS_ON (ENO, PROJNO, HOURS) - Calculate the numbers of male and female employees. - List the names of employees who are working for CSE department. - Count the number of employees who are working on JAVA project. - List the employees who are not working on any projects. Create a view to show the names of all the employees along with their department names.	10 Marks
What is a Database Trigger? Create a simple UPDATE trigger using SQL on a table EMPLOYEE.	5 Marks
What are aggregate functions? List the aggregate functions supported by SQL.	5 Marks

CSE3156 _DBMS Question Bank

Consider the relation schema given below: EMPLOYEE (EMPNO, NAME, OFFICE, AGE) BOOK (ISBN, TITLE, AUTHOR, PUBLISHER) LOAN (EMPNO, ISBN, DATE) Implement the SQL queries for the following requirements. v. Find the names of employees who have borrowed a book published by McGraw-Hill. vi. List the names of the employees who have not borrowed any books. vii. List the names of the employees who have borrowed more than two books. viii. Retrieve the total number of books present in the library. Generate a report containing EMPNO, NAME, ISBN, AUTHOR, PUBLISHER, and BORROWED DATE.	10 Marks
Explain the informal design guidelines used in relational schema design.	5 Marks
Illustrate with an example what is meant by partial functional dependency and describe how this type of dependency relates to 2NF.	5 Marks
Construct a Database to illustrate BCNF (Boyce-Codd Normal Form). How does it differ from 3NF? Explain with an example.	10 Marks
Explain the Armstrong's Rules of Inference.	5 Marks
Given a relation R(A, B, C, D) and Functional Dependency set $FD = \{ AB \rightarrow CD, B \rightarrow C \}$, determine whether the given R is in 2NF? If not convert it into 2 NF.	5 Marks
What is normalization? Design your own databases to illustrate 1NF, 2NF, and 3NF.	10 Marks
With a neat Sketch explain the different states of a database transaction.	5 Marks
Differentiate strict two-phase locking protocol and rigorous two-phase locking protocol.	5 Marks
What are concurrent transactions? Why concurrency control is required? How is it implemented in DBMS? Explain in detail.	10 Marks
List and explain the ACID properties of a database transaction.	5 Marks
Demonstrate recoverable schedule with suitable example.	5 Marks
Explain Time stamp ordering algorithm. Explain in detail the ARIES recovery method.	10 Marks
Explain the different types of DBMS users.	5 Marks
Define the terms: Database schema and Database state.	5 Marks

CSE3156 _DBMS Question Bank

Explain in detail the components of DBMS environment with a block diagram.	10 Marks																																			
List and explain various types of entity attributes with suitable examples.	5 Marks																																			
Construct E-R diagram of the bank. It provides different kinds of bank accounts. And loans. It operates number of branches.	5 Marks																																			
Consider the following relations EMPLOYEE <table><tr><th>ENO</th><th>NAME</th><th>DOB</th><th>GENDER</th><th>DCODE</th></tr><tr><td>12345</td><td>HAMEN</td><td>24-MAR-2001</td><td>M</td><td>201</td></tr><tr><td>12346</td><td>VINI</td><td>12-MAR-2001</td><td>F</td><td>202</td></tr><tr><td>12347</td><td>ANI</td><td>11-JAN-1999</td><td>F</td><td></td></tr><tr><td>12348</td><td>PETER</td><td>14-FEB-2001</td><td>M</td><td></td></tr></table> DEPARTMENT <table><tr><th>DCODE</th><th>DNAME</th></tr><tr><td>201</td><td>COMPUTER SC</td></tr><tr><td>202</td><td>INFN SC</td></tr><tr><td>203</td><td>CIVIL</td></tr><tr><td>204</td><td>MECHANICAL</td></tr></table> The primary key of each relation is underlined. Outline Cartesian product, equi-join, left outer join, right outer join and full outer join operations in relational algebra. Illustrate the above relational algebra operations with the EMPLOYEE and DEPARTMENT relations.	ENO	NAME	DOB	GENDER	DCODE	12345	HAMEN	24-MAR-2001	M	201	12346	VINI	12-MAR-2001	F	202	12347	ANI	11-JAN-1999	F		12348	PETER	14-FEB-2001	M		DCODE	DNAME	201	COMPUTER SC	202	INFN SC	203	CIVIL	204	MECHANICAL	10 Marks
ENO	NAME	DOB	GENDER	DCODE																																
12345	HAMEN	24-MAR-2001	M	201																																
12346	VINI	12-MAR-2001	F	202																																
12347	ANI	11-JAN-1999	F																																	
12348	PETER	14-FEB-2001	M																																	
DCODE	DNAME																																			
201	COMPUTER SC																																			
202	INFN SC																																			
203	CIVIL																																			
204	MECHANICAL																																			
What is a View? How can it be created and dropped? Also explain the problems associated with view update. Demonstrate with simple SQL queries.	5 Marks																																			
Explain the following SQL commands/Keywords: vi. DISTINCT vii. EXISTS LIKE	5 Marks																																			
Consider the following relations for a company database Application. EMPLOYEE (<u>ENO</u> , ENAME, SEX, DOB, DOJ, BASIC_PAY, DEPTNO) DEPARTMENT (<u>DEPTNO</u> , DNAME) PROJECT (<u>PROJNO</u> , PNAME, DEPTNO) WORKS_ON (ENO, PROJNO, HOURS) v. Calculate the numbers of male and female employees. vi. List the names of employees who are working for CSE department. vii. Count the number of employees who are working on JAVA project. viii. List the employees who are not working on any projects.	10 Marks																																			

CSE3156 _DBMS Question Bank

Create a view to show the names of all the employees along with their department names.	
What is a Database Trigger? Create a simple UPDATE trigger using SQL on a table EMPLOYEE.	5 Marks
What is a stored procedure? Create a stored procedure to add two numbers and print the result.	5 Marks
Consider the relation schema given below: EMPLOYEE (EMPNO, NAME, OFFICE, AGE) BOOK (ISBN, TITLE, AUTHOR, PUBLISHER) LOAN (EMPNO, ISBN, DATE) Implement the SQL queries for the following requirements. ix. Find the names of employees who have borrowed a book published by McGraw-Hill. x. List the names of the employees who have not borrowed any books. xi. List the names of the employees who have borrowed more than two books. xii. Retrieve the total number of books present in the library. Generate a report containing EMPNO, NAME, ISBN, AUTHOR, PUBLISHER, and BORROWED DATE.	10 Marks
Explain the informal design guidelines used in relational schema design.	5 Marks
What is a multivalued dependency? How is it different from a functional dependency? Explain with an example.	5 Marks
Explain Boyce-Codd Normal Form (BCNF). How does it different from 3NF? Explain with an example.	10 Marks
Explain the Armstrong's Rules of Inference.	5 Marks
What is a lossless decomposition? Explain with an example.	5 Marks
What is normalization? Explain 1NF, 2NF and 3NF with real world examples.	10 Marks
Explain the different states of a transaction in detail with the help of a well-labeled state diagram.	5 Marks
Explain the concept of concurrency control in transaction management. Analyze why it is essential in multi-user database environment?	5 Marks
What is 2PL? List and explain the different variations of 2PL. How does 2PL guarantee serializability? Explain with an example.	10 Marks
List and explain the ACID properties of a database transaction.	5 Marks

CSE3156 _DBMS Question Bank

Explain Timestamp ordering algorithm.	5 Marks
Explain in detail the ARIES recovery method.	10 Marks